

Scissors Excision and Other Treatment Modalities

Scissors excision is useful when smaller numbers of lesions are present and may be assisted by diathermy to control bleeding and to destroy any residual wart tissue. Clearance rates of 94–100% for electrosurgery and 89–100% for scissors excision have been reported, with recurrence rates ranging from 19–29%.

Curettage is a simple and effective technique for treating a small number of lesions, with either diathermy or silver nitrate used for hemostasis. Carbon dioxide and YAG lasers deliver very high power densities to small tissue volumes. Both electrosurgery and laser surgery should be performed using surgical masks by the treatment team and with the aid of a smoke evacuator. No suitable randomized controlled trials (RCTs) evaluating curettage or laser therapy were identified.

Formal Surgery (IVc)

Bulky warts, extensive warts, anal or intra-anal warts, and significant lesions in children are best managed under general anesthesia by an appropriate surgical specialist.

Therapies Not Generally Recommended

The routine use of **podophyllin** or **interferons** is not recommended. Podophyllin (20–25%), a non-standardized resin extract from the *Podophyllum* plant, is inexpensive but has only moderate efficacy. Up to 10% of podophyllin contains mutagenic compounds such as quercetin and kaempferol. Severe systemic toxicity following topical application has been reported, including fatalities, intrauterine death, teratogenicity, and neurological complications.

In specialist settings, α - and β -**interferons** may be used as adjuvants to surgery in difficult cases. A recent systematic review supports the use of topical or intralesional interferon. Combination therapies are commonly used in clinical practice, and further evaluation of these approaches is warranted.

Treatment Algorithms

Clinicians managing patients with genital warts should have access to both patient-applied (home) and provider-administered (clinic-based) therapies. The choice of treatment should be based on the morphology and extent of warts and made collaboratively between the clinician and patient.

Patients with limited disease (1–5 warts) often prefer immediate in-clinic treatment. Since genital warts may regress spontaneously, watchful waiting is a valid option at any anatomical site. A simple treatment algorithm is provided below.

Management of Specific Wart Locations

- **Vaginal warts:** Can be treated with trichloroacetic acid (TCA) or cryotherapy.
- **Cervical warts:** Should be managed by a gynecologist.
- **Intra-meatal warts:** Treatable with TCA or various surgical techniques.
- **Intra-anal warts:** Small-volume lesions may be treated with TCA. Imiquimod is feasible with motivated patients. Otherwise, formal surgical referral is indicated.

Treatment in Pregnancy

Genital warts may enlarge and multiply during pregnancy. The presence of warts at delivery carries an approximate risk of 1 in 400 for the infant developing juvenile-onset recurrent respiratory papillomatosis. There is no evidence that treating maternal warts reduces this risk, although reducing viral load may be prudent. Coordination with the obstetrician is recommended. Spontaneous regression of genital warts is commonly observed in the postpartum period.

Treatment in Immunocompromised Patients

Both **HIV infection** and **iatrogenic immunosuppression** are associated with higher prevalence of anogenital warts. Treatment response is often suboptimal in HIV-positive individuals, and recurrence rates are increased.

Recent studies show total clearance rates of 31–32% with imiquimod in patients on HAART. One RCT reported 100% primary clearance with a combination of surgery and imiquimod.

Partner Notification

- Current sexual partners, and if possible, partners from the past 6 months, should be evaluated for visible lesions.
- Partners should also receive education and counseling regarding sexually transmitted diseases and prevention strategies.